

Category Theory and Functional Programming

Natural Transformations

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Preliminaries

Textbook

S. Abramsky and N. Tzevelekos (2011). Introduction to Categories and Categorical Logic.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Outline

Introduction

Definition of a Natural Transformation

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Natural Isomorphisms

Natural Transformations between Hom-Functors

Compositions of Natural Transformations

Functor Category

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Definition of a Natural Transformation

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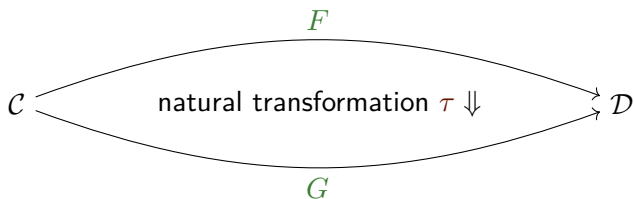
Functor Category

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Introduction

Description

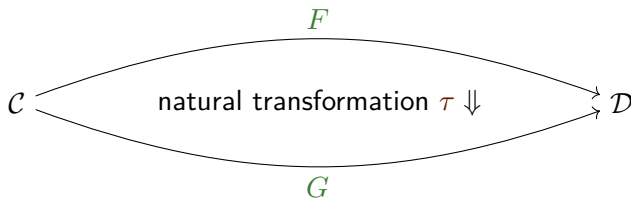
A natural transformation is a structure preserving mapping (i.e. preserves composition of arrows and identity arrows) between “parallel” functors.



Introduction

Description

A natural transformation is a structure preserving mapping (i.e. preserves composition of arrows and identity arrows) between “parallel” functors.



“As Eilenberg-Mac Lane first observed, ‘category’ has been defined in order to be able to define ‘functor’ and ‘functor’ has been defined in order to be able to define ‘natural transformation’.” (Mac Lane 1998, p. 48)

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Definition of a Natural Transformation

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be categories and let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be functors. A **natural transformation**[†]

$$\tau : F \Rightarrow G$$

is a family of arrows in $\mathcal{D}$ indexed by objects A of $\mathcal{C}$,

$$\{\tau_A : F_0 A \rightarrow G_0 A\}_{A \in \text{Obj}(\mathcal{C})} \quad (\text{components of } \tau \text{ at } A)$$

such that, for all $A \xrightarrow{f} B$ in $\mathcal{C}$,

$$(G_1 f) \circ \tau_A = \tau_B \circ (F_1 f) \quad (\text{naturality condition}).$$

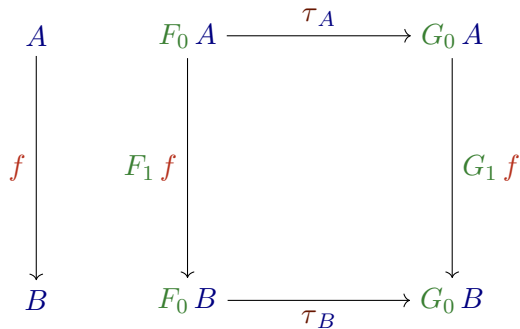
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[†]The textbook uses the notation $\tau : F \rightarrow G$.

Definition of a Natural Transformation

Definition (continuation)

That is, the following diagram commutes.



$$\left((G_1 f) \circ \tau_A = \tau_B \circ (F_1 f) \right)$$

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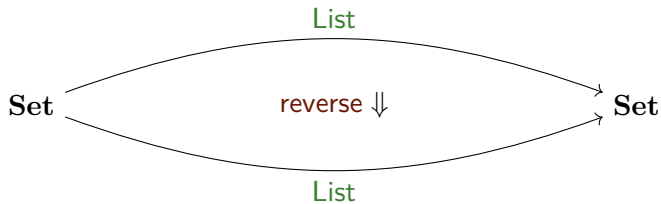
Functor Category

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Examples of Natural Transformations

Example

We shall define the natural transformation **reverse** on the functor **List** : **Set** $\rightarrow$ **Set**.



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Examples of Natural Transformations

Example (continuation)

$\text{reverse} : \text{List} \Rightarrow \text{List}$

$\text{reverse}_X : \text{List}_0 X \rightarrow \text{List}_0 X$

$\text{reverse}_X : [X] \rightarrow [X]$

$\text{reverse}_X [x_1, \dots, x_n] := [x_n, \dots, x_1]$

Examples of Natural Transformations

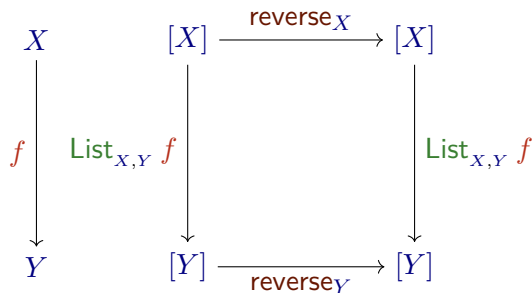
Example (continuation)

$\text{reverse} : \text{List} \Rightarrow \text{List}$

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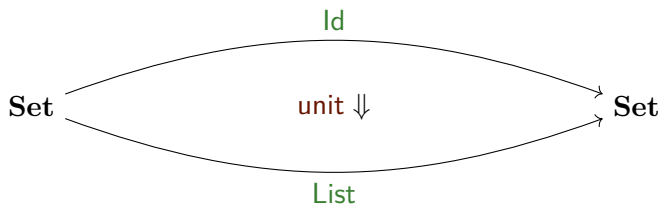


For each $f : X \rightarrow Y$ in **Set**, the above diagram commutes by naturality.

Examples of Natural Transformations

Example

We shall define the natural transformation **unit** on the functors **Id**, **List** : **Set** → **Set**.



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Examples of Natural Transformations

Example (continuation)

$\text{unit} : \text{Id} \Rightarrow \text{List}$

$\text{unit}_X : \text{Id}_0 X \rightarrow \text{List}_0 X$

$\text{unit}_X : X \rightarrow [X]$

$\text{unit}_X x := [x]$

Examples of Natural Transformations

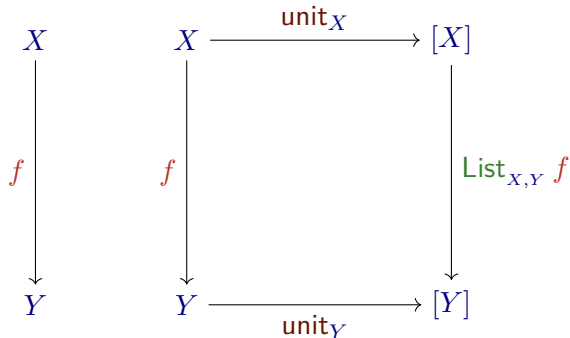
Example (continuation)

$\text{unit} : \text{Id} \Rightarrow \text{List}$

$\text{unit}_X : \text{Id}_0 X \rightarrow \text{List}_0 X$

$\text{unit}_X : X \rightarrow [X]$

$\text{unit}_X x := [x]$

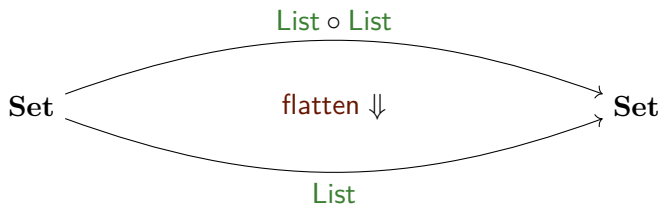


For each $f : X \rightarrow Y$ in **Set**, the above diagram commutes by naturality.

Examples of Natural Transformations

Example

We define the natural transformation **flatten** on the functors $\text{List} \circ \text{List}, \text{List} : \mathbf{Set} \rightarrow \mathbf{Set}$.



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Examples of Natural Transformations

Example (continuation)

$\text{flatten} : \text{List} \circ \text{List} \Rightarrow \text{List}$

$\text{flatten}_X : \text{List}_0 (\text{List}_0 X) \rightarrow \text{List}_0 X$

$\text{flatten}_X : [[X]] \rightarrow [X]$

$\text{flatten}_X [[x_1^1, \dots, x_{n_1}^1], \dots, [x_1^k, \dots, x_{n_k}^k]] := [x_1^1, \dots, x_{n_1}^1, \dots, x_1^k, \dots, x_{n_k}^k]$

Examples of Natural Transformations

Example (continuation)

$$\text{flatten} : \text{List} \circ \text{List} \Rightarrow \text{List}$$

$$\text{flatten}_X : \text{List}_0 (\text{List}_0 X) \rightarrow \text{List}_0 X$$

$$\text{flatten}_X : [[X]] \rightarrow [X]$$

$$\text{flatten}_X [[x_1^1, \dots, x_{n_1}^1], \dots, [x_1^k, \dots, x_{n_k}^k]] := [x_1^1, \dots, x_{n_1}^1, \dots, x_1^k, \dots, x_{n_k}^k]$$

For each $f : X \rightarrow Y$ in **Set**, the following diagram commutes by naturality.

$$\begin{array}{ccccc} X & & [[X]] & \xrightarrow{\text{flatten}_X} & [X] \\ \downarrow f & & \downarrow (\text{List} \circ \text{List})_{X,Y} f & & \downarrow \text{List}_{X,Y} f \\ Y & & [[Y]] & \xrightarrow{\text{flatten}_Y} & [Y] \end{array}$$

Examples of Natural Transformations

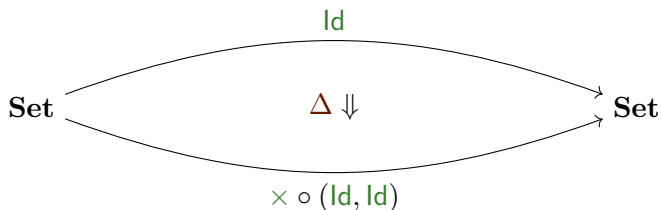
Exercise

Verify naturality of `reverse`, `unit` and `flatten`.

Examples of Natural Transformations

Example

Let Id be the identity functor in $\mathbf{Set}$ and let $\times \circ (\text{Id}, \text{Id}) : \mathbf{Set} \rightarrow \mathbf{Set}$ the functor sending every set X to $X \times X$ and every function f to $f \times f$. We shall define the natural transformation **diagonal** Δ .



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Examples of Natural Transformations

Example (continuation)

$$\Delta : \text{Id} \Rightarrow \times \circ (\text{Id}, \text{Id})$$

$$\Delta_X : \text{Id}_0 X \rightarrow (\times \circ (\text{Id}, \text{Id}))_0 X$$

$$\Delta_X : X \rightarrow X \times X$$

$$\Delta_X x := (x, x)$$

Examples of Natural Transformations

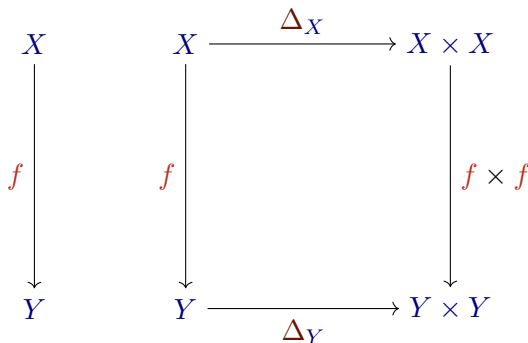
Example (continuation)

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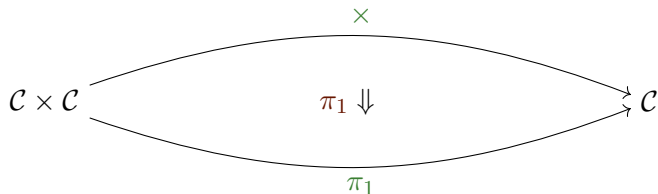


For each $f : X \rightarrow Y$ in **Set**, the above diagram commutes by naturality.

Examples of Natural Transformations

Example

Let $\times, \pi_1 : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ be the product and first projection functors, respectively. We shall define the natural transformation **first projection** π_1 .



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Examples of Natural Transformations

Example (continuation)

$$\pi_1 : \times \Rightarrow \pi_1$$

$$\pi_{1(A,B)} : \times_0(A, B) \rightarrow (\pi_1)_0(A, B)$$

$$\pi_{1(A,B)} : A \times B \rightarrow A$$

Examples of Natural Transformations

Example (continuation)

$$\pi_1 : \times \Rightarrow \pi_1$$

$$\pi_{1(A,B)} : \times_0(A, B) \rightarrow (\pi_1)_0(A, B)$$

$$\pi_{1(A,B)} : A \times B \rightarrow A$$

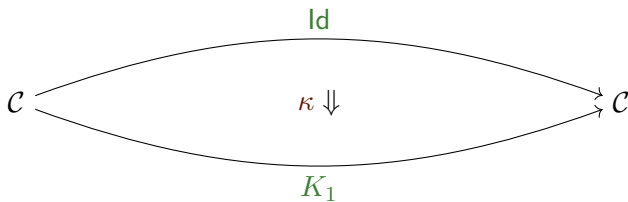
$$\begin{array}{ccccc} (A, B) & A \times B & \xrightarrow{\pi_{1(A,B)}} & A & \\ \downarrow (f, g) & \downarrow f \times g & & \downarrow f & \\ (A', B') & A' \times B' & \xrightarrow{\pi_{1(A', B')}} & A' & \end{array}$$

For each $(f, g) : (A, B) \rightarrow (A', B')$ in $\mathcal{C} \times \mathcal{C}$, the above diagram commutes by naturality.

Examples of Natural Transformations

Example

Let $\mathcal{C}$ be a category with terminal object 1 , let Id be the identity functor in $\mathcal{C}$ and let $K_1 : \mathcal{C} \rightarrow \mathcal{C}$ be the functor mapping all objects of $\mathcal{C}$ to 1 and all arrows of $\mathcal{C}$ to id_1 . We shall define the natural transformation κ .



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Examples of Natural Transformations

Example (continuation)

$$\kappa : \text{Id} \Rightarrow K_1$$

$$\kappa_A : \text{Id}_0 A \rightarrow (K_1)_0 A$$

$$\kappa_A : A \rightarrow 1$$

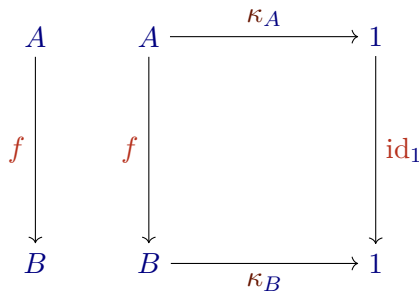
Examples of Natural Transformations

Example (continuation)

$$\kappa : \text{Id} \Rightarrow K_1$$

$$\kappa_A : \text{Id}_0 A \rightarrow (K_1)_0 A$$

$$\kappa_A : A \rightarrow 1$$



For each $f : A \rightarrow B$ in $\mathcal{C}$, the above diagram commutes by naturality.

Exercises

Exercise

Verify naturality of the natural transformation κ .

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Natural Isomorphisms

Definition

A natural transformation

$$\begin{aligned}\tau : F &\Rightarrow G \\ \tau_A : F_0 A &\rightarrow G_0 A, \quad \text{for all } A \text{ in } \mathcal{C}\end{aligned}$$

is a **natural isomorphism** if each τ_A is an isomorphism.

Natural Isomorphisms

Definition

A natural transformation

$$\tau : F \Rightarrow G$$
$$\tau_A : F_0 A \rightarrow G_0 A, \quad \text{for all } A \text{ in } \mathcal{C}$$

is a **natural isomorphism** if each τ_A is an isomorphism.

Notation

A natural isomorphism is denoted by $\tau : F \xRightarrow{\cong} G$.

Natural Isomorphisms

Example

Let $\mathcal{C}$ be a category with products and let the functors (textbook and (Awodey 2010, § 7.4))

$$-_1 \times (-_2 \times -_3) : \mathcal{C}^3 \rightarrow \mathcal{C} \quad \text{and} \quad (-_1 \times -_2) \times -_3 : \mathcal{C}^3 \rightarrow \mathcal{C}.$$

The natural isomorphism a shows that the product is associative.

$$\begin{aligned} a &: -_1 \times (-_2 \times -_3) \xrightarrow{\cong} (-_1 \times -_2) \times -_3 \\ a_{A,B,C} &: A \times (B \times C) \xrightarrow{\cong} (A \times B) \times C \\ a_{A,B,C} &:= \langle \langle \pi_1, \pi_1 \circ \pi_2 \rangle, \pi_2 \circ \pi_2 \rangle. \end{aligned}$$

Natural Isomorphisms

Example

Let $\mathcal{C}$ be a category with binary products. We define the functor $\bar{\times} : \mathcal{C}^2 \rightarrow \mathcal{C}$ by (textbook and (Awodey 2010, Example 7.8))

$$\bar{\times}_0(A, B) := \times_0(B, A) = B \times A, \quad \bar{\times}_1(f, g) := \times_1(g, f) = g \times f.$$

The natural isomorphism s shows that the product is symmetric.

$$\begin{aligned} s : \times &\xrightarrow{\cong} \bar{\times} \\ s_{A,B} : A \times B &\xrightarrow{\cong} B \times A \\ s_{A,B} &:= \langle \pi_2, \pi_1 \rangle. \end{aligned}$$

$$\begin{array}{c} (A, B) \\ \downarrow (f, g) \\ (A', B') \end{array}$$

$$\begin{array}{ccc} A \times B & \xrightarrow{s_{A,B}} & B \times A \\ \downarrow f \times g & & \downarrow g \times f \\ A' \times B' & \xrightarrow{s_{A',B'}} & B' \times A' \end{array}$$

Natural Isomorphisms

Example

Let $\mathcal{C}$ be a category with binary products and terminal object 1 and let the functors

$$1 \times - : \mathcal{C} \rightarrow \mathcal{C}$$

$$(1 \times -)_0 A := 1 \times A$$

$$(1 \times -)_1 f := \langle \text{id}_1, f \rangle,$$

$$- \times 1 : \mathcal{C} \rightarrow \mathcal{C}$$

$$(- \times 1)_0 A := A \times 1$$

$$(- \times 1)_1 f := \langle f, \text{id}_1 \rangle.$$

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Natural Isomorphisms

Example (continuation)

The natural isomorphisms l and r show that 1 is the unit of the product.

$$\begin{aligned}l &: 1 \times - \xrightarrow{\cong} \text{Id} \\l_A &: 1 \times A \xrightarrow{\cong} A \\l_A &:= \pi_2,\end{aligned}$$

$$\begin{aligned}r &: - \times 1 \xrightarrow{\cong} \text{Id} \\r_A &: A \times 1 \xrightarrow{\cong} A \\r_A &:= \pi_1.\end{aligned}$$

$$\begin{array}{ccccc}A & 1 \times A & \xrightarrow{l_A} & A \\f \downarrow & \langle \text{id}_1, f \rangle \downarrow & & \downarrow f \\B & 1 \times B & \xrightarrow{l_B} & B\end{array}$$

$$\begin{array}{ccccc}A & A \times 1 & \xrightarrow{r_A} & A \\f \downarrow & \langle f, \text{id}_1 \rangle \downarrow & & \downarrow f \\B & B \times 1 & \xrightarrow{r_B} & B\end{array}$$

Natural Isomorphisms

Exercise

Verify that the families of arrows $s_{A,B}$, l_A and r_A , from the previous examples, are natural isomorphisms (textbook, Exercise 53).

Natural Isomorphisms

Exercise

Verify that the families of arrows $s_{A,B}$, l_A and r_A , from the previous examples, are natural isomorphisms (textbook, Exercise 53).

Remark

Because natural isomorphisms are a self-dual notion, we get also natural isomorphisms s , l and r for a category with binary coproducts and initial object.

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Natural Transformations between Hom-Functors

Definition

Let $\mathcal{C}$ be a (locally small) category and let A and B be objects of $\mathcal{C}$. Recall the hom-functors

$$\mathcal{C}(A, -) : \mathcal{C} \rightarrow \mathbf{Set} \quad (\text{covariant})$$

$$\mathcal{C}(-, B) : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set} \quad (\text{contravariant})$$

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Natural Transformations between Hom-Functors

Definition (continuation, first notation)

Let $f : A \rightarrow B$ in $\mathcal{C}$. We define the natural transformation $\mathcal{C}(f, -)$ and we show its naturality condition for $h : C \rightarrow D$ in $\mathcal{C}$.

$$\mathcal{C}(f, -) : \mathcal{C}(B, -) \Rightarrow \mathcal{C}(A, -)$$

$$\mathcal{C}(f, -)_C : \mathcal{C}(B, C) \rightarrow \mathcal{C}(A, C)$$

$$\mathcal{C}(f, -)_C g : \mathcal{C}(A, C)$$

$$\mathcal{C}(f, -)_C g := g \circ f$$

$$\begin{array}{ccccc} & & \mathcal{C}(B, C) & \xrightarrow{\mathcal{C}(f, -)_C} & \mathcal{C}(A, C) \\ & & \downarrow \mathcal{C}(B, -)_1 h & & \downarrow \mathcal{C}(A, -)_1 h \\ C & \xrightarrow{h} & D & & \\ & & \mathcal{C}(B, D) & \xrightarrow{\mathcal{C}(f, -)_D} & \mathcal{C}(A, D) \end{array}$$

where $\mathcal{C}(f, -)_C = \mathcal{C}(-, C)_0 f$.

Natural Transformations between Hom-Functors

Definition (continuation, second notation)

Let $f : A \rightarrow B$ in $\mathcal{C}$. We define the natural transformation $\mathcal{C}(f, -)$ and we show its naturality condition for $h : C \rightarrow D$ in $\mathcal{C}$.

$$\mathcal{C}(f, -) : \mathcal{C}(B, -) \Rightarrow \mathcal{C}(A, -)$$

$$\mathcal{C}(f, -)_C : \mathcal{C}(B, C) \rightarrow \mathcal{C}(A, C)$$

$$\mathcal{C}(f, -)_C g : \mathcal{C}(A, C)$$

$$\mathcal{C}(f, -)_C g := g \circ f$$

$$\begin{array}{ccccc} C & & \mathcal{C}(B, C) & \xrightarrow{\mathcal{C}(f, -)_C = \mathcal{C}(f, C)} & \mathcal{C}(A, C) \\ \downarrow h & & \downarrow \mathcal{C}(B, h) & & \downarrow \mathcal{C}(A, h) \\ D & & \mathcal{C}(B, D) & \xrightarrow{\mathcal{C}(f, -)_C = \mathcal{C}(f, D)} & \mathcal{C}(A, D) \end{array}$$

Natural Transformations between Hom-Functors

Exercise

Define the natural transformation $\mathcal{C}(-, f) : \mathcal{C}(-, A) \Rightarrow \mathcal{C}(-, B)$ and verify its naturality (textbook, Exercise 55).

Yoneda Lemma

Yoneda embedding

Let $\mathcal{C}$ be a locally small category and let A and B be objects of $\mathcal{C}$. For each natural transformation between hom-functors $\tau : \mathcal{C}(A, -) \rightarrow \mathcal{C}(B, -)$, there is a unique arrow $f : B \rightarrow A$ such that

$$\tau = \mathcal{C}(f, -).$$

Yoneda Lemma

Yoneda embedding

Let $\mathcal{C}$ be a locally small category and let A and B be objects of $\mathcal{C}$. For each natural transformation between hom-functors $\tau : \mathcal{C}(A, -) \rightarrow \mathcal{C}(B, -)$, there is a unique arrow $f : B \rightarrow A$ such that

$$\tau = \mathcal{C}(f, -).$$

Remark

The Yoneda lemma is a generalisation of the Yoneda embedding.

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Vertical Composition

Definition

Let $\mathcal{C}$ be a category, let $F, G, H : \mathcal{C} \rightarrow \mathcal{D}$ be functors, and let $\tau : F \Rightarrow G$ and $\mu : G \Rightarrow H$ be natural transformations. The **vertical composition** of μ and τ is the natural transformation $\mu \circ \tau$ defined by

$$\begin{aligned} \mu \circ \tau &: F \Rightarrow H \\ (\mu \circ \tau)_A &: F_0 A \rightarrow H_0 A := \mu_A \circ \tau_A, \quad \text{for all } A \text{ in } \mathcal{C}. \end{aligned}$$

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Vertical Composition

Definition (continuation)

That is, for all $f : A \rightarrow B$ in $\mathcal{C}$ the following diagram commutes.

$$\begin{array}{ccccc} A & & F_0 A & \xrightarrow{F_1 f} & F_0 B \\ \downarrow f & & \downarrow \tau_A & & \downarrow \tau_B \\ & & G_0 A & \xrightarrow{G_1 f} & G_0 B \\ & & \downarrow \mu_A & & \downarrow \mu_B \\ & & H_0 A & \xrightarrow{H_1 f} & H_0 B \end{array} \quad \left(\begin{array}{l} (G_1 f) \circ \tau_A = \tau_B \circ (F_1 f) \\ (H_1 f) \circ \mu_A = \mu_B \circ (G_1 f) \\ (H_1 f) \circ (\mu \circ \tau)_A = (\mu \circ \tau)_B \circ (F_1 f) \end{array} \right)$$

Vertical Composition

Exercise

Let $\mathcal{C}$ be a category, let $F, G, H : \mathcal{C} \rightarrow \mathcal{D}$ be functors, and let $\tau : F \Rightarrow G$ and $\mu : G \Rightarrow H$ be natural transformations. Show that $\mu \circ \tau : F \Rightarrow H$ is a natural transformation.

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Functor Category

Introduction

Since we can define an associative composition of natural transformations and this composition has an identity natural transformation, we can define a category where the objects are functors and the arrows are natural transformations.

Functor Category

Definition

Let $\mathcal{C}$ be a small category and let $\mathcal{D}$ be an arbitrary category. The **functor category** $\text{Func}(\mathcal{C}, \mathcal{D})$ is defined by

- (i) Objects: Functors $F : \mathcal{C} \rightarrow \mathcal{D}$.
- (ii) Arrows: Natural transformations $\tau : F \Rightarrow G$.

Functor Category

Definition

Let $\mathcal{C}$ be a small category and let $\mathcal{D}$ be an arbitrary category. The **functor category** $\text{Func}(\mathcal{C}, \mathcal{D})$ is defined by

- (i) Objects: Functors $F : \mathcal{C} \rightarrow \mathcal{D}$.
- (ii) Arrows: Natural transformations $\tau : F \Rightarrow G$.
- (iii) Composition of arrows: Vertical composition of natural transformations.

Functor Category

Definition

Let $\mathcal{C}$ be a small category and let $\mathcal{D}$ be an arbitrary category. The **functor category** $\text{Func}(\mathcal{C}, \mathcal{D})$ is defined by

- (i) Objects: Functors $F : \mathcal{C} \rightarrow \mathcal{D}$.
- (ii) Arrows: Natural transformations $\tau : F \Rightarrow G$.
- (iii) Composition of arrows: Vertical composition of natural transformations.
- (iv) Identity arrow

$$\text{id}_F : F \rightarrow F$$

$$(\text{id}_F)_A : F_0 A \rightarrow F_0 A \quad \text{for all } A \text{ in } \mathcal{C},$$

$$(\text{id}_F)_A := \text{id}_{(F_0 A)}.$$

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